

Research Internship

Production and structural/photophysical analysis of layered copper silicates with differing earth alkaline metal ion contents

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April 17, 2023 - July 07, 2023

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Semester of Enrolment: FS2023

Abstract

Layered copper silicates, such as Egyptian Blue and its relatives, possess vibrant and long-lasting colors. Apart from their historical significance and artistic applications, these pigments hold promise in bioimaging, particularly in the near-infrared range, offering improved signal quality thanks to light being able to penetrate deeper into tissue due to less absorption and scattering by tissue. This work presents a comprehensive study on the synthesis and characterization of layered copper silicates with varying earth alkaline metal ion contents using flame spray pyrolysis (FSP). The objective was to investigate their crystal structure, absorption and fluorescence properties, and photoluminescence quantum yield. Annealing the copper silicate samples at 1000°C resulted in a well-resolved and crystalline structure, as confirmed by X-ray powder diffraction analysis. Luminescence measurements showed red-shifted fluorescence peaks ranging from 940 to 990 nm, with broader peaks observed at higher wavelengths. Photoluminescence quantum yield measurements revealed that annealing at 1000°C for 10 minutes yielded the highest quantum yield for most samples, with an increase observed with prolonged annealing duration. The successful incorporation of earth alkaline metals into the crystal structure of copper silicates was confirmed due to changes in structural and photonic properties. However, the incorporation of magnesium in a 1:1 calcium:magnesium ratio was unsuccessful. This study provides valuable insights into the synthesis and characterization of layered copper silicates using flame spray pyrolysis, shedding light on their crystal structure and photonic properties. The findings contribute to the potential applications of these materials in fields such as bioimaging and further investigations exploring additional earth alkaline metal ratios may lead to the discovery of compounds with improved luminescence properties.

1 Introduction

Layered copper silicates are a fascinating class of pigments that have played a significant role in the history of art and continue to captivate scientists and researchers today [1][2]. Among the most notable examples of these pigments are Egyptian Blue and Han Blue. These pigments exhibit strong color intensity, excellent lightfastness, and good chemical stability. Their layered crystal structure allows for the absorption and emission of light at specific wavelengths, resulting in vibrant and long-lasting colors [3][4]. Traditionally, these pigments were produced through complex and labor-intensive processes involving heating, grinding, and mixing specific raw materials [5]. Beyond their historical significance and artistic applications, copper silicates have shown promise in the field of bioimaging. Near-infrared (near-IR) emitting compounds pose immense potential in bioimaging because they enable deep tissue penetration and minimal background interference, resulting in improved signal quality and imaging capabilities for studying complex biological processes in vivo [6][7][8]. Near-IR light is also less phototoxic and synthetic copper silicates tend to undergo photobleaching extremely slowly, which allows for extended imaging sessions while causing less harm to tissues [3][9]. In this work, we aimed to produce layered copper silicates with differing earth alkaline metal ion contents (Ba^{2+} , Sr^{2+} , Ca^{2+} , Mg^{2+}) by means of flame spray pyrolysis (FSP). We additionally investigated the physical properties by examining the crystal structure, the absorption and fluorescence properties, and the photoluminescence quantum-yield.

2 Methods

2.1 Particle Synthesis

Precursor solutions were prepared according to Table 1. For each layered silicate compound produced, the proportions of added alkaline earth metals were changed according to the stoichiometric distribution of the desired product. Table 2 depicts all compounds produced in this work. The concentration of the desired product in the precursor amounted to 0.2M. The alkaline earth metal components were first dissolved in 2-Ethylhexanoic acid (2-EHA, 99%, Sigma-Aldrich) under reflux and heating (100-130°C) before cooling and addition of Tetrahydrofuran (THF, Sigma-Aldrich) in a 1:1 ratio between 2-EHA:THF. Finally, the precursor solution was completed by addition of the copper solution (Borchers ® Deca Copper 8) and Hexamethyldisiloxane (98%+, Alfa Aesar).

Table 1: Composition of 80 mL, 0.2 M single alkaline earth metal containing precursor solutions.

Sample	Alkaline earth metal	Borchers Deca Copper 8	Hexamethyl-disiloxane	1:1 2-EHA:THF
$\text{CaCuSi}_4\text{O}_{10}$	20 mL Ca naphthenate (Strem Chemicals)	11.2 mL	6.8 mL	42 mL
$\text{BaCuSi}_4\text{O}_{10}$	4.087 g Ba acetate (Sigma-Aldrich)	11.2 mL	6.8 mL	62 mL
$\text{SrCuSi}_4\text{O}_{10}$	3.291 g Sr acetate (Sigma-Aldrich)	11.2 mL	6.8 mL	62 mL
$\text{MgCuSi}_4\text{O}_{10}$	3.431 g Mg acetate (Sigma-Aldrich)	11.2 mL	6.8 mL	62 mL

Table 2: List of all compounds produced in this project.

$\text{Ca}_{0.5}\text{Mg}_{0.5}\text{CuSi}_4\text{O}_{10}$	$\text{Sr}_{0.33}\text{Ca}_{0.33}\text{Mg}_{0.33}\text{CuSi}_4\text{O}_{10}$
$\text{Ca}_{0.9}\text{Mg}_{0.1}\text{CuSi}_4\text{O}_{10}$	$\text{Ba}_{0.33}\text{Ca}_{0.33}\text{Mg}_{0.33}\text{CuSi}_4\text{O}_{10}$
$\text{Sr}_{0.5}\text{Mg}_{0.5}\text{CuSi}_4\text{O}_{10}$	$\text{Ba}_{0.33}\text{Sr}_{0.33}\text{Mg}_{0.33}\text{CuSi}_4\text{O}_{10}$
$\text{Ba}_{0.25}\text{Sr}_{0.25}\text{Ca}_{0.25}\text{Mg}_{0.25}\text{CuSi}_4\text{O}_{10}$	

The precursor solution was fed into a spray nozzle with a flow rate of 5 mL/min and dispersed by 5L/min oxygen. The spray was ignited while maintaining an oxygen/methane flow-ratio of 1.5/3.2 L/min. The nucleated particles were collected on a glass microfiber filter approximately 60 centimeters above the flame by means of a gas pump. During spraying, the temperature at the filter never exceeded 200°C. The particle production rate amounted to approx. 3.5g/h. The collected product was annealed in a test series at temperatures ranging from 700°C to 1100°C (100°C steps) for 10 minutes using a thermogravimetric device (Mettler Toledo, TGA/DSC 3+ Star System). Each layered silicate sample was additionally annealed at 1000°C for 24 hours in a laboratory oven (Thermolyne, Thermo Scientific).

2.2 Characterization

After annealing, the crystalline structure was analyzed by means of X-ray powder diffraction (D2 Phaser, Bruker). The intensity of diffracted X-rays was measured for 87 minutes between 2-theta[0-80°]. UV-Vis absorption measurements were taken with a (Cary 500 Scan, Vary) in a praying mantis diffuse reflectance setup. Absorbance was measured between 400-800nm in six cycles with a scan rate of 120 nm/min. Fluorescence measurements were taken with a (Spectrofluorometer FS5, Edinburgh Instruments). The powders were excited with a 750 nm laser (FC-750-5W, CNI) and detected by means of an InGaAs sensor. The emission was measured in 2 cycles between 850-1100 nm with a dwell time of 0.4 seconds. PL-QY measurements were taken with a Hamatsu Photonics Quantaaurus-QY Absolute PL quantum yield spectrometer. The powders were put into capped synthetic quartz dishes that suppress fluorescence and luminescence. The powders were excited between 500-700nm in 5nm steps.

3 Results and discussion

The X-ray diffraction (XRD) analysis was performed to investigate the crystal structure of the produced copper silicates (CS) and its dependence on the annealing procedure's duration and temperature. At lower temperatures (>800°C), the CS primarily exhibited an amorphous state, evidenced by an unwell-resolved diffraction pattern in the XRD spectrum. However, annealing at 1000°C resulted in a more ordered and crystalline structure, as evidenced by distinct and characteristic peaks in the XRD spectra (see Fig. 1A). All CS samples followed a similar pattern in the XRD spectra. To assess the similarity of the annealed CS, their XRD spectra were compared to a reference of $\text{CaCuSi}_4\text{O}_{10}$. Although slight angular shifts ($\pm 0.5\text{-}2^\circ$) were observed in all spectra, the main crystal phases remained similar to the reference sample. Notably, the $\text{Ca}_{0.5}\text{Mg}_{0.5}\text{CuSi}_4\text{O}_{10}$ sample displayed a distinct XRD spectrum that fully deviated from the reference spectra (see Fig. 1B).

UV-VIS absorption measurements provided insight into the optical properties of the copper silicate samples. With the exception of $\text{Ca}_{0.5}\text{Mg}_{0.5}\text{CuSi}_4\text{O}_{10}$, $\text{Ca}_{0.9}\text{Mg}_{0.1}\text{CuSi}_4\text{O}_{10}$ (1000°C) and $\text{Ba}_{0.33}\text{Ca}_{0.33}\text{Mg}_{0.33}\text{CuSi}_4\text{O}_{10}$, all CS samples exhibited an absorption peak around 630nm, accompanied by a shoulder at around 550nm. $\text{Ca}_{0.5}\text{Mg}_{0.5}\text{CuSi}_4\text{O}_{10}$ did not exhibit any absorption between

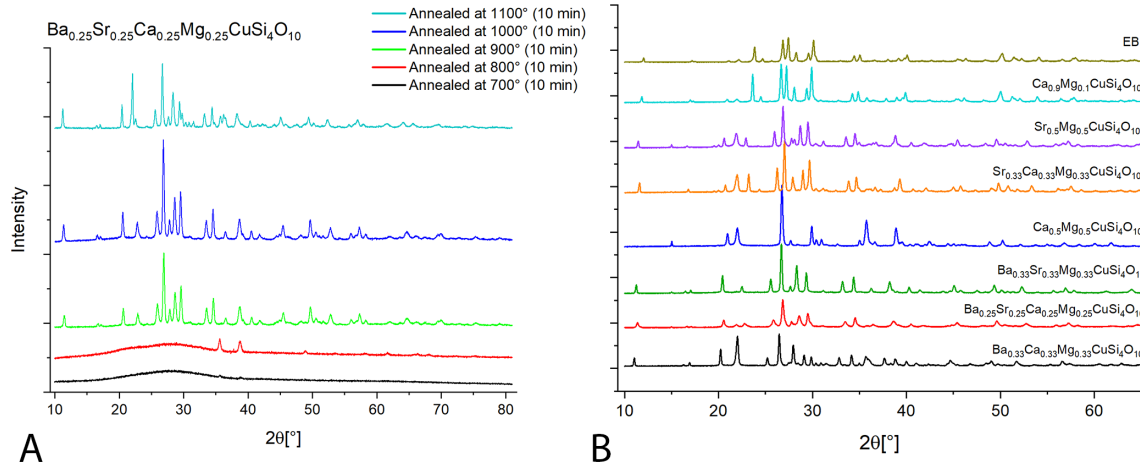


Figure 1: A) Exemplary XRD spectra of $\text{Ba}_{0.25}\text{Sr}_{0.25}\text{Ca}_{0.25}\text{Mg}_{0.25}\text{CuSi}_4\text{O}_{10}$ annealed at different temperatures for 10 minutes. B) XRD spectra of all copper silicates produced after annealing at 1000°C for 24 hours, together with an XRD spectrum of Egyptian Blue (1000°C , 24 hours) as reference.

400 and 800 nm, while $\text{Ba}_{0.33}\text{Ca}_{0.33}\text{Mg}_{0.33}\text{CuSi}_4\text{O}_{10}$ had higher absorbance at wavelengths <630 nm, accompanied by a drop in absorbance at wavelengths >630 nm. One could also very well observe, how the absorption increases around 630 nm when $\text{Ca}_{0.9}\text{Mg}_{0.1}\text{CuSi}_4\text{O}_{10}$ was annealed at 1100°C (see Fig. 2A). Regarding luminescence, the annealed CS samples (1000°C , 24h) displayed luminescence peaks ranging from 940 to 990 nm.

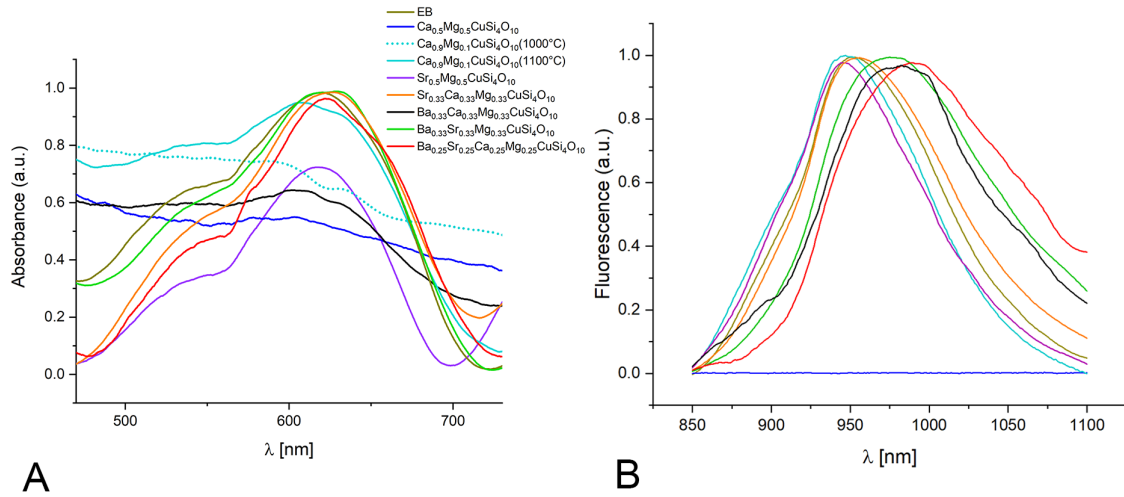


Figure 2: A) Absorption spectra between 400 and 800 nm. B) Fluorescence spectra between 850 and 1100 nm.

$\text{Ba}_{0.33}\text{Ca}_{0.33}\text{Mg}_{0.33}\text{CuSi}_4\text{O}_{10}$, $\text{Ba}_{0.33}\text{Sr}_{0.33}\text{Mg}_{0.33}\text{CuSi}_4\text{O}_{10}$ and $\text{Ba}_{0.25}\text{Sr}_{0.25}\text{Ca}_{0.25}\text{Mg}_{0.25}\text{CuSi}_4\text{O}_{10}$ exhibited the highest red shift compared to the reference sample (EB), while $\text{Ca}_{0.9}\text{Mg}_{0.1}\text{CuSi}_4\text{O}_{10}$ and $\text{Sr}_{0.5}\text{Mg}_{0.5}\text{CuSi}_4\text{O}_{10}$ showed a slight blue shift compared to EB. Additionally, the red-shifted CS samples showed broader peaks, particularly at higher wavelengths (see Fig. 2B).

Photoluminescence quantum yield (PL-QY) measurements were conducted to evaluate the differences in quantum yield based on annealing temperature and duration. In all cases, the PL-QY peaked at 1000°C annealing temperature, except for $\text{Ca}_{0.9}\text{Mg}_{0.1}\text{CuSi}_4\text{O}_{10}$ and $\text{Ca}_{0.5}\text{Mg}_{0.5}\text{CuSi}_4\text{O}_{10}$ (see Fig. 3A). Notably, $\text{Ca}_{0.9}\text{Mg}_{0.1}\text{CuSi}_4\text{O}_{10}$ exhibited its PL-QY peak at 1100°C annealing temperature when comparing annealing times of 10 minutes. When sintering for 24 hours, the same sample had its QY maximum at 1000°C, whilst decaying after sintering for 24 hours at 1100°C. The influence of annealing duration was further highlighted by comparing the PL-QY of CS annealed at 1000°C for 10 minutes and 24 hours. This comparison revealed a significant increase in PL-QY with prolonged annealing duration, except for $\text{Ca}_{0.5}\text{Mg}_{0.5}\text{CuSi}_4\text{O}_{10}$ (see Fig. 3B).

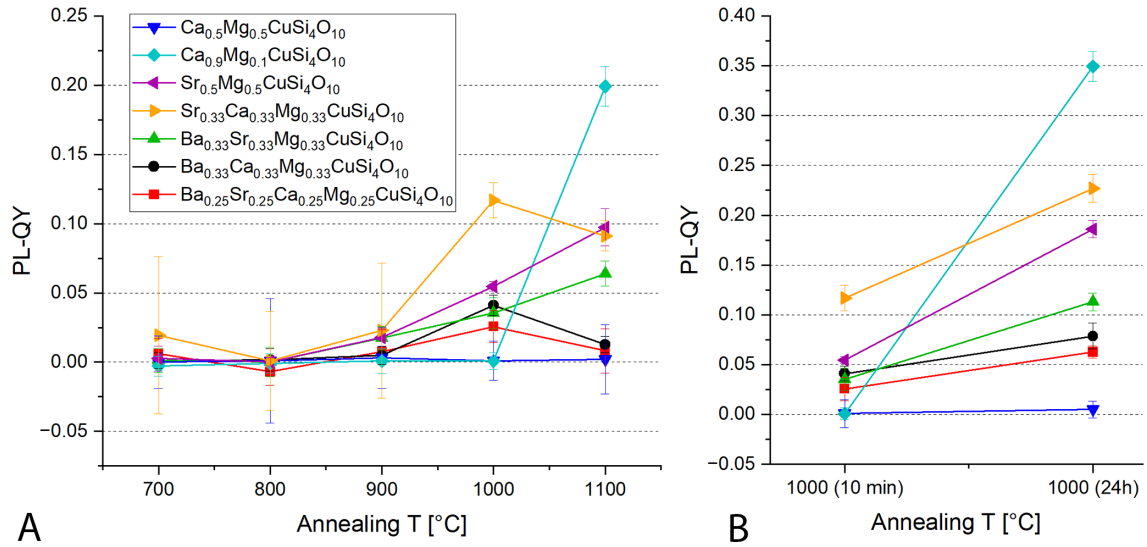


Figure 3: A) Evolution of PL-QY at different annealing temperatures. B) Comparison of PL-QY between annealing times of 10 minutes and 24 hours.

The characterization techniques employed in this project confirmed the successful incorporation of earth alkaline metals (Ba, Sr, Ca, Mg) into the crystal structure typically found in Egyptian Blue. However, the incorporation of magnesium in a 1:1 calcium/magnesium ratio proved to be unsuccessful, as indicated by incomparable XRD spectra, the absence of an absorption peak at 630 nm, and the lack of luminescence at 950 nm. Nevertheless, successful incorporation of magnesium in EB crystal structure was achieved at lower magnesium ratios. The photoluminescence quantum yield (PL-QY) of $\text{Ca}_{0.9}\text{Mg}_{0.1}\text{CuSi}_4\text{O}_{10}$ (annealed at 1000°C for 24 hours) exceeded 30%, however the luminescence peak exhibited only a minimal shift of 5 nm towards the blue compared to EB.

The XRD analysis provided insights into the crystal structure of the copper silicates produced via flame spray pyrolysis. It was observed that the sintering temperature significantly influenced the crystal structure and thus changed the photonic properties. An annealing temperature of 1000°C was proven to be the most favorable. At lower temperatures, the compound remained amorphous, while at higher temperatures (1100°C), the compound was destroyed. Absorption measurements further

confirmed that all copper silicate samples exhibited absorption peaks in the range of 630 to 640 nm, consistent with previous literature on EB’s absorbance properties [3][10][11]. The luminescence peaks of the copper silicate samples were observed around 950 nm, which is slightly higher than emission maxima reported in literature (910-940 nm) [11][12]. However, deviations from the EB reference peak (between 940-990 nm) indicated successful incorporation of earth alkaline metals, resulting in slight alterations to the crystalline and photoluminescent properties.

The PL-QY analysis demonstrated that the quantum yield could be tailored by adjusting the annealing time and temperature. Optimal quantum yield was achieved at an annealing temperature of 1000°C. Furthermore, the results revealed that prolonged annealing led to higher quantum yields. This controllability and adjustability of the crystal structure and photoluminescent properties through annealing time and temperature hold significant promise for customization based on specific requirements. Compounds such as BaCaSrMg exhibited a significant luminescence shift towards higher wavelengths, which could potentially enhance tissue penetrability. However, this red shift was accompanied by a decrease in quantum yield, diminishing the bioimaging potential. It is important to note that only a limited number of earth alkaline metal ratios were tested, suggesting that further exploration of different compositions may yield even more promising luminescence properties.

This study demonstrated the successful incorporation of earth alkaline metals in varying compositions of copper silicates. The manipulation of annealing time and temperature allowed control over the crystal structure and photoluminescent properties. The findings highlight the potential of these compounds, offering adjustability and controllability for various applications. The red shifted compounds have yet to be tested in order to see if they possess any advantages to e.g. EB in terms of tissue penetrability and imageability. Further investigations exploring additional earth alkaline metal ratios may lead to the discovery of improved luminescence properties, expanding the possibilities for bioimaging applications such as tissue imaging.

4 Conclusion

This study successfully developed a robust protocol for the synthesis of copper silicates with different earth alkaline metal contents using flame spray pyrolysis. By manipulating the annealing temperature and duration, the crystalline and photoluminescent properties of the copper silicates were effectively adjusted. The results demonstrated that longer annealing times led to higher quantum yields, indicating improved luminescent properties. Furthermore, an annealing temperature of 1000°C was found to yield an optimal crystal structure, resulting in enhanced luminescence characteristics. Despite variations in emission properties among the compounds, all produced copper silicates exhibited the typical absorption properties of EB, with an absorption peak at 630 to 640 nm. In conclusion, copper silicates with varying earth alkaline metal contents exhibited versatile near-IR luminescence properties. These nanoparticles hold significant potential for applications in bioimaging, such as single particle tracking in vivo. Future research in this field could further explore the specific applications and potential benefits of these copper silicate nanoparticles in biomedical imaging and related fields. Further investigations exploring additional earth alkaline metal ratios may lead to the discovery of improved luminescence properties, expanding the possibilities for bioimaging applications.

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